

Deploy to Dev — `appleboy/scp-action` SSH Authentication Failure

1. Summary

The **Deploy — Development** workflow's `deploy-dev` job fails at the "Copy artefact to Dev server" step (`appleboy/scp-action@v1`). This is a missing/misconfigured **GitHub Environment secret**, not a code or workflow-logic bug — no source file change can fix it. It must be resolved in the repository's GitHub settings by someone with admin access.

2. Where it happens

- **Workflow:** `.github/workflows/deploy.yml`
- **Job:** `deploy-dev` (`environment: development`)
- **Step:** `Copy artefact to Dev server` (`uses: appleboy/scp-action@v1`)
- **Trigger:** any push to `main` (the job runs after `build-and-test` and `publish` succeed)

The same secret set is also consumed by:

- `.github/workflows/release.yml` → `deploy-dev` job (identical secrets, triggered on version tags)
- `.github/workflows/release.yml` → `deploy-uat` job (equivalent `UAT_*` secrets)
- `.github/workflows/release.yml` → `deploy-prod` job (equivalent `PROD_*` secrets)

So this is not a one-off — if `DEV_SSH_KEY` is missing, **every workflow that deploys to Dev** will fail the same way, and the equivalent `UAT_SSH_KEY` / `PROD_SSH_KEY` secrets should be checked too before they're needed.

3. Raw error

```
Run appleboy/scp-action@v1
Run echo "$GITHUB_ACTION_PATH" >> $GITHUB_PATH
Run entrypoint.sh
Downloading drone-scp-1.8.0-linux-amd64 from https://github.com/appleboy/drone-scp/releases/download/v1.8.0
===== CLI Version Information =====
Drone SCP version 1.8.0
=====
2026/07/06 02:07:40 Error: can't connect without a private SSH key or password
Error: Process completed with exit code 1.
```

4. Root cause

`appleboy/scp-action` (and the matching `appleboy/ssh-action` step right after it) requires either a `key` (private SSH key) or a `password` input to authenticate to the target host. In `deploy.yml`:

```
- name: Copy artefact to Dev server
  uses: appleboy/scp-action@v1
  with:
    host: ${ secrets.DEV_HOST }
    username: ${ secrets.DEV_USER }
    key: ${ secrets.DEV_SSH_KEY }
    port: ${ secrets.DEV_PORT }
    source: api.zip
    target: /tmp/
```

`key: ${ secrets.DEV_SSH_KEY }` resolved to an **empty string**, so the action had nothing to authenticate with and drone-scp refused to even attempt a connection. GitHub Actions silently resolves an undefined/unset secret to `""` rather than failing the workflow at expansion time, which is why this only surfaces here, deep in the third-party action's own validation.

This means one of the following is true in the repo's GitHub settings:

- `DEV_SSH_KEY` was never created, or
- `DEV_SSH_KEY` exists at the wrong scope (e.g. a repository secret when the job pins `environment: development`, or vice versa — environment secrets and repository secrets do not automatically fall back to one another), or
- `DEV_SSH_KEY` exists but is empty/blank.

5. Required secrets (full picture)

Environment	Secrets required	Used by
development	DEV_HOST, DEV_USER, DEV_SSH_KEY, DEV_PORT, DEV_CONNECTION_STRING, DEV_JWT_SECRET	deploy.yml → deploy-dev, release.yml → deploy-dev
uat	UAT_HOST, UAT_USER, UAT_SSH_KEY, UAT_PORT, UAT_CONNECTION_STRING, UAT_JWT_SECRET	release.yml → deploy-uat
production	PROD_HOST, PROD_USER, PROD_SSH_KEY, PROD_PORT, PROD_CONNECTION_STRING, PROD_JWT_SECRET	release.yml → deploy-prod

Only `*_HOST`, `*_USER`, `*_SSH_KEY`, `*_PORT` are needed to fix *this specific* SCP/SSH error.

`*_CONNECTION_STRING` and `*_JWT_SECRET` are consumed later in the same job (written into `appsettings.Production.json` on the target server) — worth confirming they're populated too so the fix doesn't just move the failure one step further down the job.

6. How to fix

This must be done by someone with **admin access to the GitHub repository** (`Trust-Internship/trust-digital-communication`).

1. **Generate a dedicated deploy keypair** (don't reuse a personal SSH key):

```
ssh-keygen -t ed25519 -C "github-actions-deploy-dev" -f dev_deploy_key -N ""
```

This produces `dev_deploy_key` (private) and `dev_deploy_key.pub` (public).

2. **Authorize the public key on the Dev server**, under the account the workflow will log in as (`DEV_USER`):

```
ssh dev-server "mkdir -p ~/.ssh && cat >> ~/.ssh/authorized_keys" < dev_deploy_key.pub
```

Verify permissions are correct on the server: `~/.ssh` = 700, `authorized_keys` = 600.

3. **Add the private key as a GitHub Environment secret**, not a repository secret, since `deploy-dev` pins `environment: development`:
 - GitHub → repo → **Settings** → **Environments** → **development** → **Environment secrets** → **Add secret**
 - Name: `DEV_SSH_KEY`
 - Value: the full contents of `dev_deploy_key` (the private key file), including the `-----BEGIN OPENSSH PRIVATE KEY----- / -----END OPENSSH PRIVATE KEY-----` lines, no trailing modifications.
4. **While in that Environment secrets page, confirm the other three are also present and correct:** `DEV_HOST`, `DEV_USER`, `DEV_PORT` (typically 22 unless the server uses a non-standard port).
5. **Delete the local private key file** (`dev_deploy_key`) once it's pasted into GitHub — don't leave deploy keys sitting on a laptop or commit them anywhere in the repo.
6. Repeat steps 1–5 for `uat` (`UAT_*`) and `production` (`PROD_*`) environments before those deploy stages are exercised, to avoid hitting the identical failure later in the pipeline.

7. How to verify the fix

- Re-run the failed workflow run from the GitHub Actions UI ("Re-run failed jobs"), or push a new commit to `main` to trigger `deploy.yml` fresh.
- The `Copy artefact to Dev server` step should complete without the `drone-scp` error.
- The following step, `Deploy on Dev server` (`appleboy/ssh-action`), uses the **same** `DEV_SSH_KEY` — if step 6 succeeds, step 7 should authenticate the same way. If it fails separately, re-check that the public key was appended to the correct user's `authorized_keys` on the Dev server (not a different user than `DEV_USER`).

8. Related, but distinct, CI issue

This is unrelated to the `ConnectionStrings 'DefaultConnection' not found` failure fixed separately in `deploy.yml`'s/`release.yml`'s `build-and-test` job (see `docs/HANDOVER.md` §5's note on `ConnectionStrings__DefaultConnection`, and the fix commit adding that env var). That earlier issue was a

missing **plain env var** consumed directly by the .NET app's own configuration binding during test startup. This issue is a missing **GitHub Environment secret** consumed by a third-party SSH action in a later, separate job (**deploy-dev**) that only runs after **build-and-test** already passes. Fixing one does not fix the other.